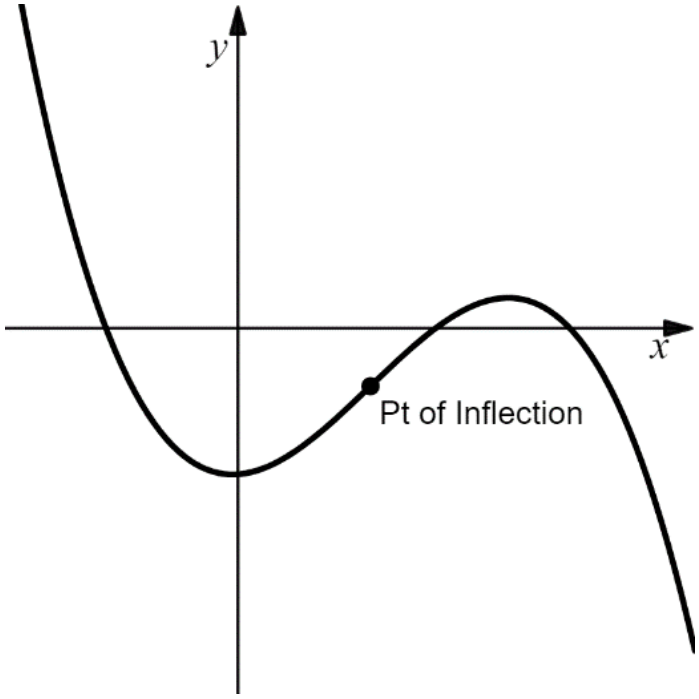
2.20 Differentiation: Points of Inflection and Applications

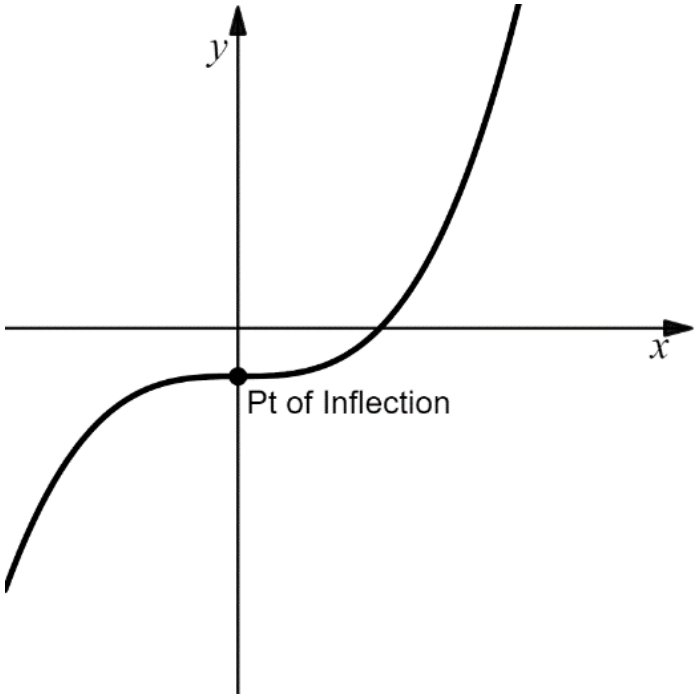
Worked Solutions

1 a)



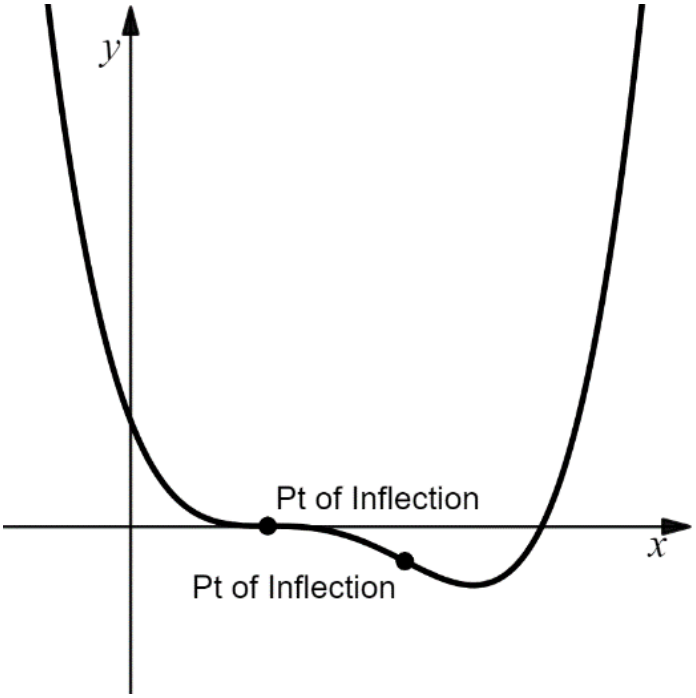
From convex (concave upwards) to concave (concave downwards)

1 b)



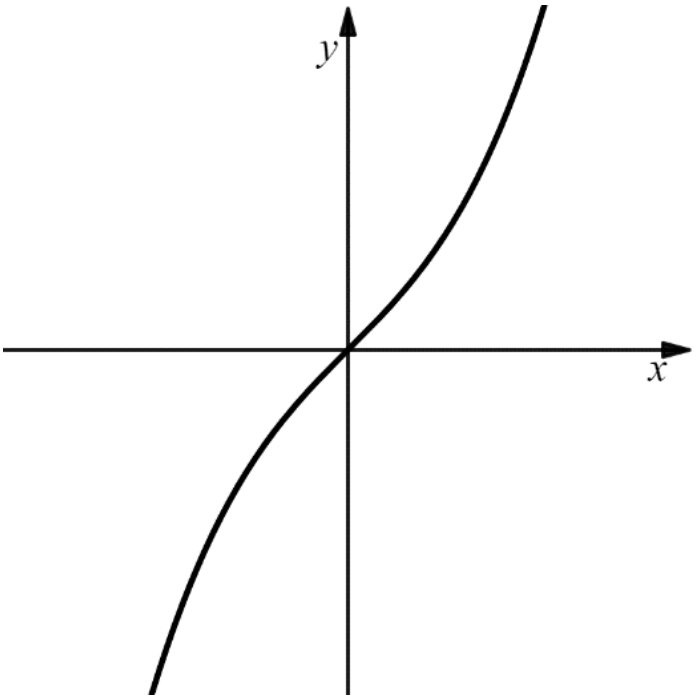
From concave (concave downwards) to convex (concave upwards)

1 c)

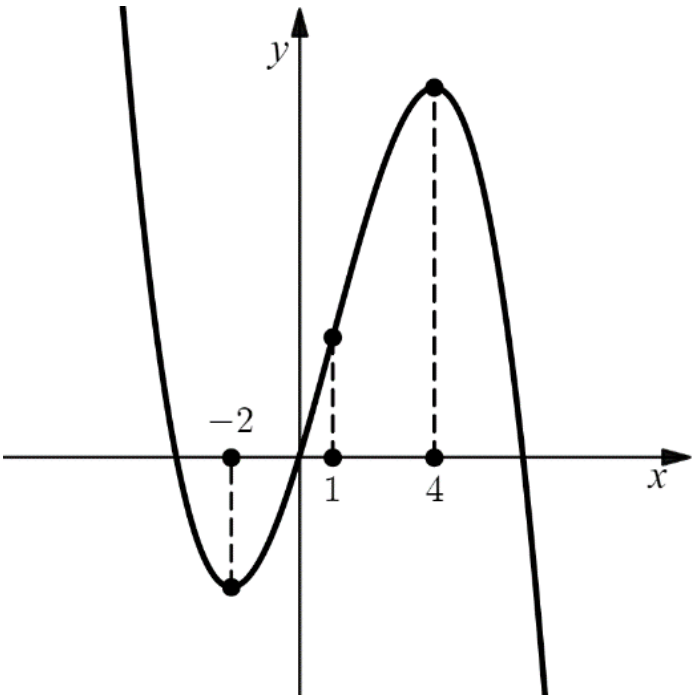


From convex (concave upwards) to concave (concave downwards) to convex (concave upwards)

2)



3)



4 a)

$$\frac{dy}{dx} = 27x^2 - 2x - 1$$

$$\frac{d^2y}{dx^2} = 54x - 2$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=0} = -2 < 0 \ \therefore \text{concave (concave downwards) at } x = 0$$

4 b)

$$\frac{dy}{dx} = 4x^3 + 6x + 1$$

$$\frac{d^2y}{dx^2} = 12x^2 + 6$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=1} = 18 > 0 \ \therefore \text{convex (concave upwards) at } x = 1$$

4 c)

$$\frac{dy}{dx} = 15x^2 - 2x + 1$$

$$\frac{d^2y}{dx^2} = 30x - 2$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-1} = -32 < 0 \ \therefore \text{concave (concave downwards) at } x = -1$$

4 d)

$$\frac{dy}{dx} = 3x^2 - 24x + 48$$

$$\frac{d^2y}{dx^2} = 6x - 24$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=4} = 0 \ \therefore \text{ neither at } x = 4$$

5)

$$\mathrm{f}'(x) = 6x^5 - 15x^4 - 40x^3 + 30x^2 + 90x$$

$$\mathrm{f}''(x) = 30x^4 - 60x^3 - 120x^2 + 60x + 90$$

Points of inflection when $\mathrm{f}''(x) = 0$

$$30x^4 - 60x^3 - 120x^2 + 60x + 90 = 0$$

$$\Rightarrow 30(x^4 - 2x^3 - 4x^2 + 2x + 3) = 0$$

$$\Rightarrow 30(x-3)(x-1)(x+1)^2 = 0$$

So possible points of inflection at $x = 3, x = 1, x = -1$

$\mathrm{f}(3) = -135, \mathrm{f}(1) = 43$ and $\mathrm{f}(-1) = 29$

x		-1		1		3	
$\mathrm{f}''(x)$	$+$	0	$+$	0	$-$	0	$+$

$\therefore$ points of inflection at only $(1, 43)$ and $(3, -135)$ as the second derivative does not change either side of $(-1, 29)$ (this is known as a point of undulation).

6)

$$\frac{dy}{dx} = 60x^4 - 80x^3 - 150x^2 + 180x$$

$$\frac{d^2y}{dx^2} = 240x^3 - 240x^2 - 300x + 180$$

Convex (concave upwards) when $\frac{d^2y}{dx^2} > 0$

$$240x^3 - 240x^2 - 300x + 180 > 0$$

$$4x^3 - 4x^2 - 5x + 3 > 0$$

$$(2x-1)(2x-3)(x+1) > 0$$

So $-1 < x < \frac{1}{2}$ and $x > \frac{3}{2}$

$$\left\{x : x \in \mathbb{R}, -1 < x < \frac{1}{2}\right\} \cup \left\{x : x \in \mathbb{R}, x > \frac{3}{2}\right\}$$

7 a)

$$\mathrm{f}'(x) = 4x^3 + 27x^2 - 30x$$
$$\mathrm{f}''(x) = 12x^2 + 54x - 30$$

Point of inflection when $\mathrm{f}''(x) = 0$

$$12x^2 + 54x - 30 = 0$$
$$\Rightarrow 6(2x^2 + 9x - 5) = 6(2x - 1)(x + 5) = 0$$

So $x = -5$ and $x = \frac{1}{2}$

$\mathrm{f}(-5) = -871$ and $\mathrm{f}(\frac{1}{2}) = \frac{23}{16}$

so points of inflection at $(-5, -871)$ and $(\frac{1}{2}, \frac{23}{16})$

$\mathrm{f}'(-5) = 325$ so not stationary

$\mathrm{f}'(\frac{1}{2}) = -\frac{31}{4}$ so not stationary

7 b)

$$\mathrm{f}'(x) = 4x^3 + 6x^2 - 2$$
$$\mathrm{f}''(x) = 12x^2 + 12x$$

Point of inflection when $\mathrm{f}''(x) = 0$

$$12x^2 + 12x = 0$$
$$\Rightarrow 12x(x + 1) = 0$$

So $x = 0$ and $x = -1$

$\mathrm{f}(0) = -1$ and $\mathrm{f}(-1) = 0$

so points of inflection at $(0, -1)$ and $(-1, 0)$

$\mathrm{f}'(0) = -2$ so not stationary

$\mathrm{f}'(-1) = 0$ so stationary

Non-stat pt of inflection at $(0, -1)$

Stat pt of inflection at $(-1, 0)$

8)

$$\frac{dh}{dt} = 6t^2 - 18t + 12$$
$$\frac{d^2h}{dt^2} = 12t - 18$$
$$\frac{d^2h}{dt^2} = 0$$
$$0 = 12t - 18 \ \therefore t = 1.5 \text{ seconds}$$

The rate of change in height (vertical velocity) is at its steepest between the local maximum value and the local minimum value when $t = 1.5$ seconds.

9 a)

$$V = 0.25 \times 12^3 - 3.25 \times 12^2 + 10 \times 12 + 20$$
$$= 104$$

£104

9 b)

$$\frac{dV}{dt} = 0.75t^2 - 6.5t + 10$$
$$\frac{d^2V}{dt^2} = 1.5t - 6.5$$
$$\frac{d^2h}{dt^2} = 0$$
$$0 = 1.5t - 6.5 \ \therefore t = \frac{13}{3} = 4.33 \text{ seconds (3 SF)}$$

The rate of change in the value of the shares is at its steepest between the local maximum value and the local minimum value when $t = 4.33$ months.

10)

$$\frac{dy}{dx} = 6x^2 + 32x + 43 \text{ and } \frac{d^2y}{dx^2} = 12x + 32$$

Point of inflection when $\frac{d^2y}{dx^2} = 0$

$$12x + 32 = 0 \text{ so } x = -\frac{32}{12} = -\frac{8}{3}$$

$$\left.\frac{dy}{dx}\right|_{x=-\frac{8}{3}} = \frac{1}{3} \neq 0 \text{ so not stationary.}$$

Only point of inflection is $(-\frac{8}{3}, -\frac{22}{27})$

11 a)

$\mathrm{f}'(x) = 3x^2 + 6x + 3$ stat pt when $\mathrm{f}'(x) = 0$

$$3(x^2 + 2x + 1) = 3(x + 1)^2 = 0$$

so $x = -1$ (repeated stat point)

$\mathrm{f}(-1) = -1 + 3 - 3 + 2 = 1$

so $(-1, 1)$ is the only stationary point

11 b)

$\mathrm{f}''(x) = 6x + 6$ so $\mathrm{f}''(-1) = 0$

$$\left.\frac{dy}{dx}\right|_{x=-1.1} = 0.03 \text{ and } \left.\frac{dy}{dx}\right|_{x=-0.9} = 0.03$$

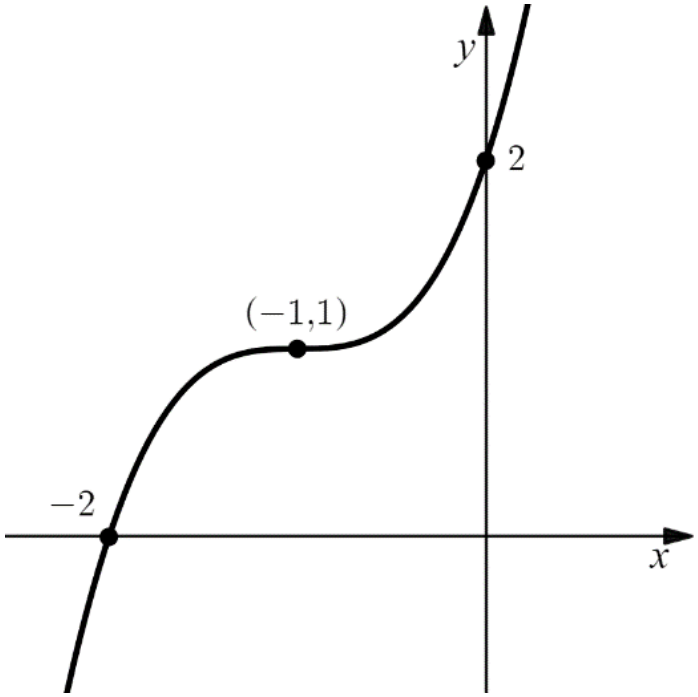
$\therefore (-1, 1)$ is a stationary point of inflection

11 c)

When $x = 0, y = 2$

When $y = 0, x^3 + 3x^2 + 3x + 2$

$= (x + 2)(x^2 + x + 1)$ so $x = -2$



12 a)

$\mathrm{f}'(x) = 2x - 3 + x^{-2}$ stat pt when $\mathrm{f}'(x) = 0$

$$2x - 3 + \frac{1}{x^2} = 0 \Rightarrow 2x^3 - 3x^2 + 1 = 0$$

$$\Rightarrow (x - 1)^2(2x + 1) = 0$$

so $x = 1$ (repeated stat point) or $x = -\frac{1}{2}$

$\mathrm{f}(1) = 1 - 3 - 1 = -3$ so $(1, -3)$

$\mathrm{f}(-\frac{1}{2}) = \frac{1}{4} + \frac{3}{2} + 2 = \frac{15}{4}$ so $(-\frac{1}{2}, \frac{15}{4})$

$\mathrm{f}''(x) = 2 - 2x^{-3}$ so $\mathrm{f}''(1) = 0$

$\mathrm{f}'(0.9) = 0.0345679$ and $\mathrm{f}'(1.1) = 0.0264463$

$\therefore (1, -3)$ is a stationary point of inflection

$\mathrm{f}''(\frac{1}{2}) = 1 - 3 + \frac{1}{4} = \frac{7}{4} > 0$ $(-\frac{1}{2}, \frac{15}{4})$ is a minimum

12 b)

$\mathrm{f}''(x) = 2 - 2x^{-3}$ pt of inflection when $\mathrm{f}''(x) = 0$

$2 - 2x^{-3} = 0 \Rightarrow 2 = \frac{2}{x^3} \Rightarrow x^3 = 1$ so pt of inflection is at $x = 1$ which is already identified as stationary.

13 a)

$f'(x) = 12x^3 - 12x^2$ stat pt when $f'(x) = 0$

$12x^3 - 12x^2 = 0 \Rightarrow 12x^2(x-1) = 0$

so $x = 0$ (repeated stat pt) and $x = 1$

$f(0) = 2$ and $f(1) = 3 - 4 + 2 = 1$

so $(0, 2)$ and $(1, 1)$ are stationary points.

13 b)

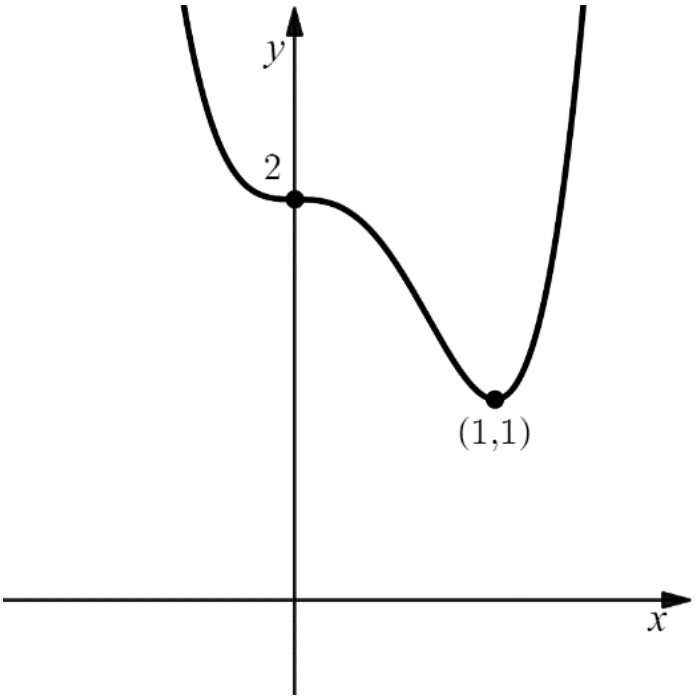
$f''(x) = 36x^2 - 24x$ so $f''(0) = 0$

$f'(-0.1) = -0.132$ and $f'(0.1) = -0.108$

So $(0, 2)$ is a stationary point of inflection.

$f''(1) = 36 - 24 = 12 > 0$ so $(1, 1)$ is a minimum

13 c)



14 a)

$f'(x) = 8x^3 + 12x^2$ stat pt when $f'(x) = 0$

$8x^3 + 12x^2 = 0 \Rightarrow 4x^2(2x+3) = 0$

so $x = 0$ (repeated stat pt) and $x = -\frac{3}{2}$

$f(0) = 0$ and $f(-\frac{3}{2}) = -\frac{27}{8}$

so $(0, 0)$ and $(-\frac{3}{2}, -\frac{27}{8})$ are stationary points.

$f''(x) = 24x^2 + 24x$ so $f''(0) = 0$

$f'(-0.1) = 0.112$ and $f'(0.1) = 0.128$

So $(0, 0)$ is a stationary point of inflection.

$f''(-\frac{3}{2}) = 18 > 0$ so $(-\frac{3}{2}, -\frac{27}{8})$ is a minimum

14 b)

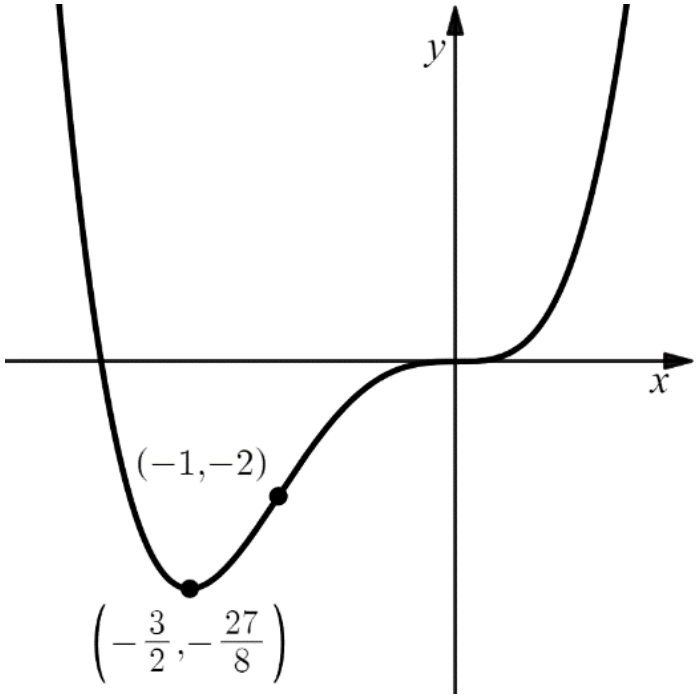
Point of inflection when $f''(x) = 0$

$24x^2 + 24x = 0 \Rightarrow 24x(x+1) = 0$ so $x = 0$ or $x = -1$

So non-stationary point of inflection is when $x = -1$

$f(-1) = -2$ that is $(-1, -2)$

14 c)



15)

$$\frac{dy}{dx} = 3x^2 + 2ax + b$$

$$\frac{d^2y}{dx^2} = 6x + 2a$$

Stationary point of inflection:

$\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} = 0$ for the same point.

At $(5, 2)$:

$\frac{d^2y}{dx^2} = 30 + 2a = 0$ so $a = -15$

$\frac{dy}{dx} = 75 - 150 + b = 0$ so $b = 75$

$2 = 125 - 375 + 375 + c$ so $c = -123$

16)

$$\frac{dy}{dx} = -3x^2 + 2ax + b$$

$$\frac{d^2y}{dx^2} = -6x + 2a$$

Stationary point of inflection:

$\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} = 0$ for the same point.

At $(1, 6)$:

$\frac{d^2y}{dx^2} = -6 + 2a = 0$ so $a = 3$

$\frac{dy}{dx} = -3 + 6 + b = 0$ so $b = -3$

$6 = -1 + 3 - 3 + c$ so $c = 7$

17 a)

$$\frac{dy}{dx} = e^x - 1 \text{ and } \frac{d^2y}{dx^2} = e^x$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=2} = e^2 > 0 \ \therefore \text{convex at } x = 2$$

17 b)

$$\frac{dy}{dx} = 2e^x - 2e^{2x} \text{ and } \frac{d^2y}{dx^2} = 2e^x - 4e^{2x}$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=0} = 2 - 4 = -2 < 0 \ \therefore \text{concave at } x = 0$$

17 c)

$$\frac{dy}{dx} = 2e^{2x} \text{ and } \frac{d^2y}{dx^2} = 4e^{2x}$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-1} = 4e^{-2} > 0 \ \therefore \text{convex at } x = 1$$

17 d)

$$\frac{dy}{dx} = -e^{-x} + 2x \text{ and } \frac{d^2y}{dx^2} = e^{-x} + 2$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-1} = e + 2 > 0 \ \therefore \text{convex at } x = -1$$

17 e)

$$\frac{dy}{dx} = -e^{-x} - 2x \text{ and } \frac{d^2y}{dx^2} = e^{-x} - 2$$
$$\left.\frac{d^2y}{dx^2}\right|_{x=0} = -1 < 0 \ \therefore \text{concave at } x = 0$$

18 a)

$$\frac{dy}{dx} = e^x - 2x \text{ and } \frac{d^2y}{dx^2} = e^x - 2$$

Point of inflection when $\frac{d^2y}{dx^2} = 0$

$$e^x - 2 = 0 \Rightarrow e^x = 2 \Rightarrow x = \ln 2$$
$$\left.\frac{dy}{dx}\right|_{x=-0.1+\ln 2} = 0.623380474952$$
$$\left.\frac{dy}{dx}\right|_{x=0.1+\ln 2} = 0.624047475031$$

So a point of inflection.

18 b)

$$y = 2 - (\ln 2)^2 \ \therefore (\ln 2, 2 - (\ln 2)^2)$$

19 a)

$$\frac{dy}{dx} = 1 + \frac{1}{2}\cos\left(\frac{1}{2}x\right)$$
$$\frac{d^2y}{dx^2} = -\frac{1}{4}\sin\left(\frac{1}{2}x\right)$$
$$-\frac{1}{4}\sin\left(\frac{1}{2}x\right) = 0 \text{ has solution } x = 2\pi$$

(in the range $0 < x \leq 2\pi$)

At $x = 2\pi$, $\frac{dy}{dx} = 1 + \frac{1}{2} \neq 0$

So there is a single non-stationary point of inflection.

19 b)

$$\frac{dy}{dx} = 2x + \frac{1}{x}$$
$$\frac{d^2y}{dx^2} = 2 - \frac{1}{x^2}$$
$$2 - \frac{1}{x^2} = 0 \Rightarrow 2 = \frac{1}{x^2} \Rightarrow x^2 = \frac{1}{2} \text{ has solution } x = \frac{1}{\sqrt{2}}$$

(in the range $x > 0$)

At $x = \frac{1}{\sqrt{2}}$, $\frac{dy}{dx} = 2 \times \frac{1}{\sqrt{2}} + \sqrt{2} \neq 0$

So there is a single non-stationary point of inflection.

20 a)

Min when $\frac{dP}{dt} = 0$

$$\begin{aligned}\frac{dP}{dt} &= 0.5(t-3)e^{0.5t} + e^{0.5t} = e^{0.5t}(0.5(t-3)+1) \\ &= e^{0.5t}(0.5t - 0.5) = 0.5e^{0.5t}(t-1)\end{aligned}$$

$\therefore t = 1$ month

$P(1) = 144 + (1-3)e^{0.5} = 140.7$ pence

20 b)

Pt of inflection when $\frac{d^2P}{dt^2} = 0$

$$\begin{aligned}\frac{d^2P}{dt^2} &= 0.5e^{0.5t} + 0.25e^{0.5t}(t-1) \\ &= e^{0.5t}(0.5 + 0.25(t-1)) \\ &= e^{0.5t}(0.5 + 0.25t - 0.25) = e^{0.5t}(0.25 + 0.25t) \\ &= 0.25e^{0.5t}(1+1t)\end{aligned}$$

$\therefore t = -1$ which is not valid for this model

According to the model, there is no point in the 12 months where there is a change in convexity of the curve. This means that the rate in change in the price of petrol decreases in the first month, before increasing for the remainder of the year.